

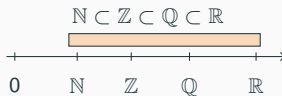
Math Bootcamp

Day 1 — Foundations (Sections 1.1–1.6)

Math Camp 2026

1.1 Number systems — names and definitions

Symbol	English name	What it is
$\mathbb{N}$	natural numbers	Non-negative integers for counting
$\mathbb{Z}$	integers	$\dots, -2, -1, 0, 1, 2, \dots$
$\mathbb{Q}$	rational numbers	Fractions p/q ($q \neq 0$)
$\mathbb{R}$	real numbers	All points on the number line
$\mathbb{R}^n$	n -dimensional real space	Lists / vectors of length n



1.1 Symbols and vectors — names and definitions

Common symbols	$\in$	“is an element of”	$\forall$	“for all”
	$\exists$	“there exists”	$\Rightarrow$	“implies”
			$\Leftrightarrow$	“if and only if” (iff)

Scalar vs. vector

- **Scalar:** one number, e.g. income $w \in \mathbb{R}$
- **Vector:** a list of numbers $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$
- **Dot product:** $\mathbf{p} \cdot \mathbf{x} = \sum_{i=1}^n p_i x_i$

1.1 — Which set? Together

Smallest set that fits?

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2. Money you spend at a vending machine (e.g. HK\$8.50)

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→ $\mathbb{Q}$

e.g. $8.50 = 17/2$ dollars

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e.g. $8.50 = 17/2$ dollars

3. Order as a vector: (HK\$ spent on drink, # snacks) = (8.50, 2)

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e.g. $8.50 = 17/2$ dollars

3. Order as a vector: (HK\$ spent on drink, # snacks) = (8.50, 2)

→ $(8.50, 2) \in \mathbb{Q} \times \mathbb{N}$

1.2 Logic — names and definitions

English	Symbol	Meaning
P only if Q	$P \Rightarrow Q$	Q is necessary for P (without Q , no P)
P if Q	$Q \Rightarrow P$	Q is sufficient for P (Q alone is enough)
P if and only if Q	$P \Leftrightarrow Q$	Q is necessary and sufficient
Contrapositive	$\neg Q \Rightarrow \neg P$	same truth as $P \Rightarrow Q$: if Q fails, P must fail

1.1 — IKEA run Together

First week in HK: you go to IKEA and spot three plush toys.



Otter HK\$179.9



Hedgehog HK\$79.9



BLÅHAJ HK\$179.9

You have **HK\$500** pocket money ($B = 500$).

Basket $\mathbf{x} \in \mathbb{N}^3$: # otters, # hedgehogs, # sharks. E.g. $(1, 1, 0)$ = one otter, one hedgehog, no shark. 6/20
Price vector $\mathbf{p} = (179.9, 79.9, 179.9) \in \mathbb{Q}^3$. Budget set: $\mathcal{B} = \{\mathbf{x} \in \mathbb{N}^3 : \mathbf{p} \cdot \mathbf{x} \leq 500\}$.

1.2 — Plush symbols practice Together

$\mathbf{p} = (179.9, 79.9, 179.9)$, $\mathcal{B} = \{\mathbf{x} \in \mathbb{N}^3 : \mathbf{p} \cdot \mathbf{x} \leq 500\}$. True or false?

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5. $[\Rightarrow]$ $(1, 1, 1) \in \mathcal{B}$ **only if** $\mathbf{p} \cdot (1, 1, 1) \leq 500$

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6. $[\Rightarrow]$ $(0, 0, 3) \in \mathcal{B}$ **if** $(1, 1, 1) \in \mathcal{B}$

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3. $[\forall]$ $\forall \mathbf{x} \in \mathcal{B}, \mathbf{x} \neq (0, 1, 0)$ **False** ($(0, 1, 0) \in \mathcal{B}$)
4. $[\exists]$ $\exists \mathbf{x} \in \mathcal{B}, \mathbf{x} = (3, 0, 0)$ **False** ($539.7 > 500$)
5. $[\Rightarrow]$ $(1, 1, 1) \in \mathcal{B}$ only if $\mathbf{p} \cdot (1, 1, 1) \leq 500$ **True** ($439.7 \leq 500$)

also True: $(1, 1, 1) \in \mathcal{B}$ if $\mathbf{p} \cdot (1, 1, 1) \leq 500$

6. $[\Rightarrow]$ $(0, 0, 3) \in \mathcal{B}$ if $(1, 1, 1) \in \mathcal{B}$ **False** ($(1, 1, 1) \in \mathcal{B}$ but $539.7 > 500$)

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6. $[\Rightarrow]$ $(0, 0, 3) \in \mathcal{B}$ if $(1, 1, 1) \in \mathcal{B}$ **False** ($(1, 1, 1) \in \mathcal{B}$ but $539.7 > 500$)
7. [contrapositive] If $\mathbf{p} \cdot (1, 1, 1) + \mathbf{p} \cdot (0, 1, 0) > 500$, then $(1, 1, 1) \notin \mathcal{B}$ or $(0, 1, 0) \notin \mathcal{B}$

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6. $[\Rightarrow] (0, 0, 3) \in \mathcal{B}$ **if** $(1, 1, 1) \in \mathcal{B}$ **False** ($(1, 1, 1) \in \mathcal{B}$ but $539.7 > 500$)
7. [contrapositive] If $\mathbf{p} \cdot (1, 1, 1) + \mathbf{p} \cdot (0, 1, 0) > 500$, then $(1, 1, 1) \notin \mathcal{B}$ **or** $(0, 1, 0) \notin \mathcal{B}$ **False** ($519.6 > 500$, yet each basket alone is in $\mathcal{B}$)

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7. [contrapositive] **If** $\mathbf{p} \cdot (1, 1, 1) + \mathbf{p} \cdot (0, 1, 0) > 500$, **then** $(1, 1, 1) \notin \mathcal{B}$ **or** $(0, 1, 0) \notin \mathcal{B}$ **False** ($519.6 > 500$, yet each basket alone is in $\mathcal{B}$)

each affordable $\nRightarrow$ jointly affordable

1.3 Sets and operations — names and definitions

Symbol	Name	Meaning
$\in, \notin$	element / not in	x belongs (or not) to A
$\subset, =$	subset, equality	$A \subset B$; $A = B$ iff mutual $\subset$
$\emptyset, \{ \}, \{x \in S : \dots\}$	sets	empty, roster, set-builder
$A \cup B, A \cap B$	union, intersection	in A or B ; in both
$A^c, A \setminus B$	complement, difference	not in A ; in A not B
$A \times B$	Cartesian product	pairs (a, b) , $a \in A$, $b \in B$

In economics: budget set $\mathcal{B} \subset \mathbb{N}^3$; $(8.50, 2) \in \mathbb{Q} \times \mathbb{N}$.

1.3 — Set operations (IKEA) Together

$\mathcal{B} = \{\mathbf{x} \in \mathbb{N}^3 : \mathbf{p} \cdot \mathbf{x} \leq 500\}$, $\mathcal{H} = \{\mathbf{x} \in \mathbb{N}^3 : \text{middle entry} \geq 1\}$ (at least one hedgehog).

Describe the set.

1.3 — Set operations (IKEA) Together

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Describe the set.

1. $\mathcal{B} \cap \mathcal{H}$

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Describe the set.

1. $\mathcal{B} \cap \mathcal{H}$ affordable baskets with ≥ 1 hedgehog (e.g. $(0, 1, 0)$)

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1. $\mathcal{B} \cap \mathcal{H}$ affordable baskets with ≥ 1 hedgehog (e.g. $(0, 1, 0)$)
2. $\mathcal{B} \cup \mathcal{H}$

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Describe the set.

1. $\mathcal{B} \cap \mathcal{H}$ affordable baskets with ≥ 1 hedgehog (e.g. $(0, 1, 0)$)
2. $\mathcal{B} \cup \mathcal{H}$ affordable baskets or baskets with ≥ 1 hedgehog

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3. $\mathcal{B} \setminus \mathcal{H}$

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2. $\mathcal{B} \cup \mathcal{H}$ affordable baskets **or** baskets with ≥ 1 hedgehog
3. $\mathcal{B} \setminus \mathcal{H}$ affordable baskets with **no** hedgehog (e.g. $(1, 0, 0)$)

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4. $(179.9, 79.9) \in \mathbb{Q} \times \mathbb{Q}$ but $(179.9, 79.9) \notin \mathcal{B}$

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4. $(179.9, 79.9) \in \mathbb{Q} \times \mathbb{Q}$ but $(179.9, 79.9) \notin \mathcal{B}$ **True** Cartesian product $\neq$ budget set — pairs $\neq$ baskets

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5. $(1, 0, 0) \subset \mathcal{B}$

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3. $\mathcal{B} \setminus \mathcal{H}$ affordable baskets with no hedgehog (e.g. $(1, 0, 0)$)
4. $(179.9, 79.9) \in \mathbb{Q} \times \mathbb{Q}$ but $(179.9, 79.9) \notin \mathcal{B}$ **True** Cartesian product $\neq$ budget set — pairs $\neq$ baskets
5. $(1, 0, 0) \subset \mathcal{B}$ **False** $(1, 0, 0)$ is a basket: use $(1, 0, 0) \in \mathcal{B}$ (True); but $\{(1, 0, 0)\} \subset \mathcal{B}$ is True

1.4 Functions — names and definitions

Term	Notation	Meaning
Function	$f: A \rightarrow B$	each $x \in A$ maps to exactly one $f(x) \in B$
Domain	A	allowed inputs
Codomain	B	target set — every output must lie in B
Range	$f(A)$	outputs hit: $\{f(x) : x \in A\} \subseteq B$
Injective (one-to-one)		$f(x_1) = f(x_2) \Rightarrow x_1 = x_2$
Surjective (onto)		every $y \in B$ equals some $f(x)$
Operations	$(f \pm g)(x)$, $(cf)(x)$, $(fg)(x)$	$(f \pm g)(x) = f(x) \pm g(x)$; $(cf)(x) = c f(x)$; $(fg)(x) = f(x)g(x)$
Composition	$(f \circ g)(x)$	apply g first: $(f \circ g)(x) = f(g(x))$

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Define $f: D \rightarrow C$ by: $f(x)$ = your girlfriend / boyfriend's age when **your age** is x .

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7. If $f(25) = \{22, 80\}$, is f a function? **No — a correspondence** (a function gives **exactly one** output per input; it's immoral to have a correspondence ☺)

1.5 — Reservation price (example first) Together

BLÅHAJ sticker price: **HK\$179.9**. You buy **only on sale**: when $p < 179.9$.

Prices where you still buy: $P = \{p \in \mathbb{R}_+ : p < 179.9\}$.

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Reservation price (WTP)? $\sup P = 179.9$ least upper bound — the choke price)

Is $179.9 \in P$? **No** at sticker price you walk away $\Rightarrow$ no max P)

1.5 Supremum and infimum — names and definitions

Term	Notation	Meaning
Maximum	$\max S$	largest element in S
Minimum	$\min S$	smallest element in S
Supremum	$\sup S$	least upper bound of S
Infimum	$\inf S$	greatest lower bound of S

Example (least upper bound): $S = [0, 1)$. Upper bounds include $1, 2, 3, \dots$; $\sup S = 1$ — the **smallest** upper bound ($1 \notin S$).

Relations: if $\max S$ exists, $\max S = \sup S$; if $\min S$ exists, $\min S = \inf S$.

Always: $\inf S \leq x \leq \sup S$ for all $x \in S$.

1.6 Taylor, logs, and exponentials

Taylor expansion (around a):

$$f(a+h) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} h^k = f(a) + f'(a)h + \frac{f''(a)}{2!}h^2 + \frac{f^{(3)}(a)}{3!}h^3 + \dots$$

Logs

$$\ln(ab) = \ln a + \ln b$$

$$\ln(x^\alpha) = \alpha \ln x$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$$

Exponentials

$$e^{a+b} = e^a e^b$$

$$e^{\alpha x} = (e^x)^\alpha$$

$$e^x = 1 + x + \frac{x^2}{2!} + \dots$$

Quick check: $\ln(x_1^\alpha x_2^\beta) = \alpha \ln x_1 + \beta \ln x_2$; if $u = x_1^\alpha x_2^{1-\alpha}$, then $\ln u = \alpha \ln x_1 + (1-\alpha) \ln x_2$.

1.6 Why logs?

- **Taylor:** $f(a + h) \approx f(a) + f'(a)h$.
- **Apply to logs:** $\ln(1 + x) \approx x$ for small x .
- **Percentage changes:** $\ln(x + \Delta x) - \ln x = \ln\left(1 + \frac{\Delta x}{x}\right) \approx \frac{\Delta x}{x}$, so $\Delta \ln x$ tracks % change in x .
- **Example:** $\frac{\Delta x}{x} = 5\% \Rightarrow \Delta \ln x \approx 0.05$.
- **Log-log regression:** $\ln y = \alpha + \beta \ln x$.
- $1\% \uparrow$ in $x \Rightarrow \beta\%$ change in y .
- **Elasticity:** $\varepsilon_{y,x} = \frac{\partial y}{\partial x} \frac{x}{y} = \frac{\Delta y/y}{\Delta x/x} = \frac{\partial \ln y}{\partial \ln x}$; in $\ln y = \alpha + \beta \ln x$, $\varepsilon_{y,x} = \beta$.

1.6 Cobb–Douglas — definition

Utility / production:

$$U(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}, \quad Y(K, L) = AK^\alpha L^{1-\alpha}, \quad \alpha \in (0, 1).$$

In logs (linear): $\ln U = \alpha \ln x_1 + (1 - \alpha) \ln x_2$; $\ln Y = \ln A + \alpha \ln K + (1 - \alpha) \ln L$.

Constant elasticity: $\varepsilon_{U, x_1} = \frac{\partial \ln U}{\partial \ln x_1} = \alpha$, $\varepsilon_{U, x_2} = 1 - \alpha$; $\varepsilon_{Y, K} = \alpha$, $\varepsilon_{Y, L} = 1 - \alpha$. Same numbers at every (x_1, x_2) or (K, L) .

1.6 Why Cobb–Douglas? — first-order approximation

Any smooth utility $u(x_1, x_2) > 0$: define $f(\ln x_1, \ln x_2) = \ln u(x_1, x_2)$.

First-order Taylor around $(\ln \bar{x}_1, \ln \bar{x}_2)$:

$$\ln u(x_1, x_2) \approx f(\ln \bar{x}_1, \ln \bar{x}_2) + \left. \frac{\partial f}{\partial \ln x_1} \right|_{\bar{x}} (\ln x_1 - \ln \bar{x}_1) + \left. \frac{\partial f}{\partial \ln x_2} \right|_{\bar{x}} (\ln x_2 - \ln \bar{x}_2).$$

Let $\alpha_i = \left. \frac{\partial \ln u}{\partial \ln x_i} \right|_{\bar{x}}$ and absorb constants into c :

$$\ln u(x_1, x_2) \approx c + \alpha_1 \ln x_1 + \alpha_2 \ln x_2 \quad \Leftrightarrow \quad u(x_1, x_2) \approx e^c x_1^{\alpha_1} x_2^{\alpha_2}.$$

Near $\bar{x}$ only; Cobb–Douglas is exact when $\ln u$ is globally affine in logs.

1.6 Cobb–Douglas — homogeneity and CRS

Homogeneous of degree k : $f(tx) = t^k f(x)$ for $t > 0$.

Utility $U = x_1^\alpha x_2^{1-\alpha}$: $U(tx_1, tx_2) = t^\alpha t^{1-\alpha} U(x) = t U(x) \Rightarrow$ degree 1.

Production $Y = AK^\alpha L^{1-\alpha}$: $Y(tK, tL) = t^\alpha t^{1-\alpha} Y(K, L) = t Y(K, L)$.

CRS (constant returns to scale): double all inputs $\Rightarrow$ double output (homogeneous of degree 1).

1.6 Cobb–Douglas — constant expenditure shares

$$\text{Max } U = x_1^\alpha x_2^{1-\alpha} \text{ s.t. } p_1 x_1 + p_2 x_2 = w.$$

$$\text{FOC (marginal utility per dollar equal): } \frac{\partial U / \partial x_1}{\partial U / \partial x_2} = \frac{p_1}{p_2} \Rightarrow \frac{\alpha/x_1}{(1-\alpha)/x_2} = \frac{p_1}{p_2} \Rightarrow \frac{p_1 x_1}{p_2 x_2} = \frac{\alpha}{1-\alpha}.$$

$$\text{Budget binds: } p_1 x_1 + p_2 x_2 = w. \text{ Substitute } p_1 x_1 = \frac{\alpha}{1-\alpha} p_2 x_2:$$

$$w = p_2 x_2 \left(\frac{\alpha}{1-\alpha} + 1 \right) = \frac{p_2 x_2}{1-\alpha} \Rightarrow p_2 x_2 = (1-\alpha)w, \quad p_1 x_1 = \alpha w.$$

$$\text{Expenditure shares: } \frac{p_1 x_1}{w} = \alpha, \quad \frac{p_2 x_2}{w} = 1 - \alpha \text{ (constant).}$$

1.6 Why Cobb–Douglas? — summary

Why economists use Cobb–Douglas:

- **Logs = % changes:** elasticities are $\partial \ln y / \partial \ln x$.
- **Local approximation:** any smooth $u > 0$ satisfies $\ln u \approx c + \alpha_1 \ln x_1 + \alpha_2 \ln x_2$ near $\bar{\mathbf{x}}$.
- **Constant elasticity / shares:** $\varepsilon_{u, x_i} = \alpha_i$; CD demands give fixed budget shares.
- **Homogeneity (degree 1) / CRS:** $f(t\mathbf{x}) = tf(\mathbf{x})$; double inputs $\Rightarrow$ double output.
- **Easy algebra:** linear in logs, powers in levels; closed-form demands and FOCs.
- **Global use = modeling choice:** tractability and homogeneity, not a theorem for all preferences.